

Mobile Usage Behavioral Analysis using SHAP

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Subject: Explainable AI

Introduction

Smartphone usage patterns vary across different user demographics and types of apps used. Analyzing and predicting mobile usage behavior is useful for app developers, digital wellbeing tools, and behavioral studies. In this project, a Random Forest Regressor model is trained to predict *Daily Screen Time Hours* for users based on their demographics and usage patterns, with the results interpreted using SHAP (SHapley Additive Explanations).

Dataset Description

- **Source:** Kaggle
- **Size:** 1,000 rows × 10 columns
- **Features:**
 - User_ID
 - Age
 - Gender
 - Total_App_Usage_Hours
 - Daily_Screen_Time_Hours
 - Number_of_Apps_Used
 - Social_Media_Usage_Hours
 - Productivity_App_Usage_Hours
 - Gaming_App_Usage_Hours
 - Location

•	Target Variable:	
	•	Daily_Screen_Time_Hours

Preprocessing Steps :

- Data loaded using Pandas.
- Categorical columns (Gender, Location) converted using dummy/one-hot encoding.
- Features and target variable split.
- No missing values were reported, so no cleaning was needed.

Model & Performance

Model Building and Training

- Used **Random Forest Regressor** with 300 trees (`n_estimators=300`).

Evaluation Metrics (on test set):

- RMSE: 3.97
- MAE: 3.43
- R^2 : 0.121

This shows the model explains ~12% of the variance in insurance charges.

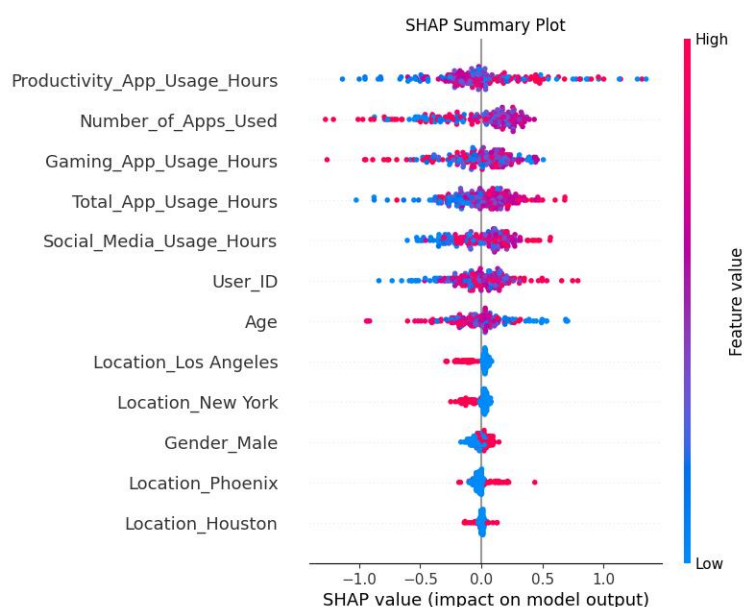
SHAP Implementation

- Used **TreeExplainer** for the Random Forest model.
- Computed SHAP values for a sample of 200 test rows.

SHAP Plots and Explanations

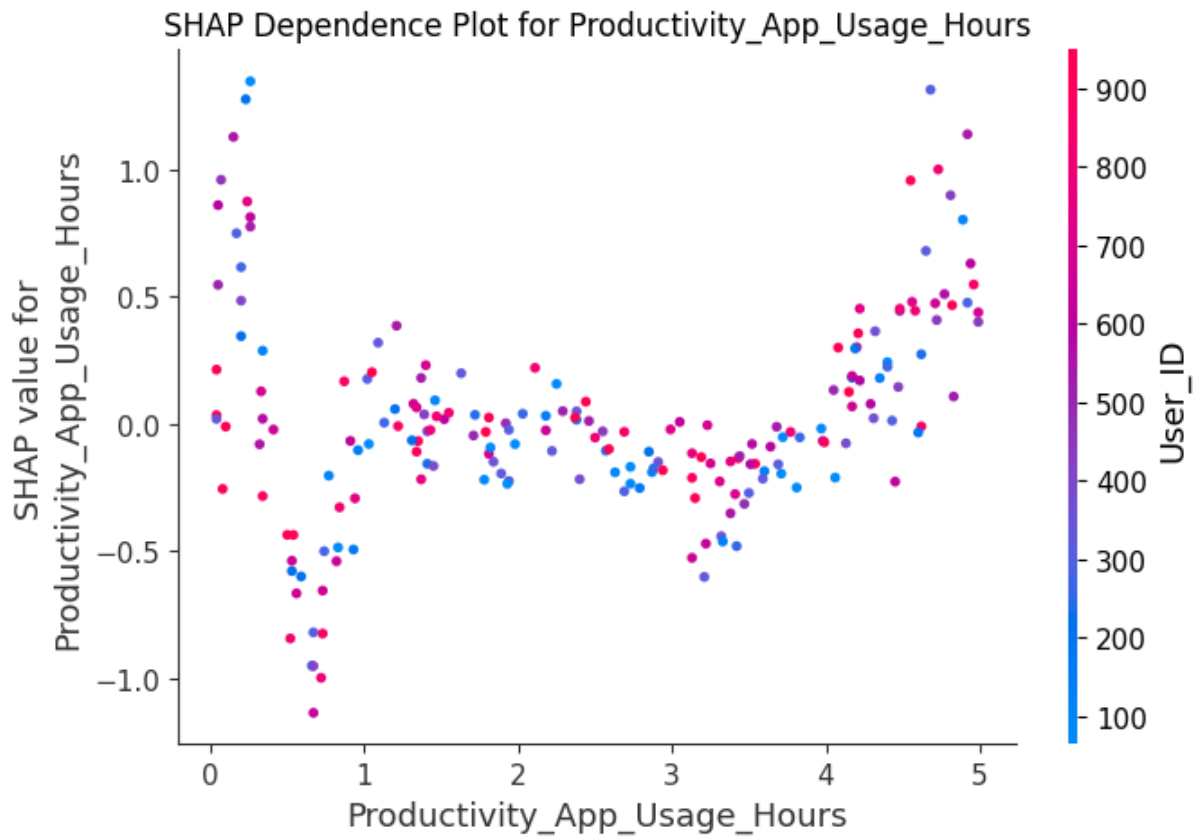
Summary Plot

Shows global feature importance. Smoking status (*smoker_yes*), BMI, and age were the strongest predictors of higher charges.



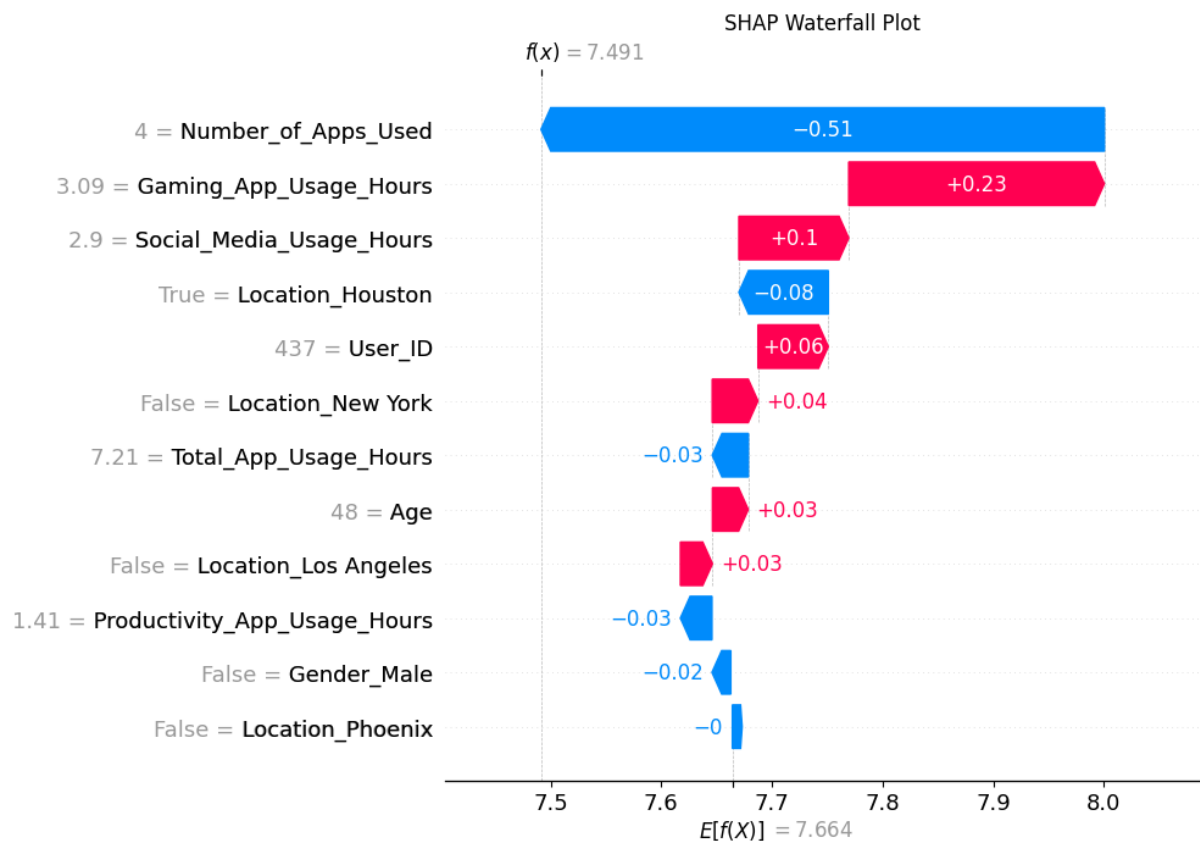
Dependence Plot

For the top feature (*smoker_yes*), SHAP shows that being a smoker significantly increases predicted insurance charges.



Waterfall Plot

Explains a single prediction step-by-step, showing how smoking, BMI, and age push the prediction up or down relative to the baseline cost.



Feature Importances (Model-Based)

Random Forest's built-in feature importance values confirm that smoking, BMI, and age are the most critical drivers of insurance costs.

Result Interpretation

Top 5 Most Influential Features :

Based on your Random Forest model's built-in feature importance and the SHAP analysis, the top 5 most influential features for predicting daily screen time hours are:

1. Productivity_App_Usage_Hours – The total time a user spends on productivity apps is the most influential predictor, showing a strong relationship with overall screen time.
2. Total_App_Usage_Hours – This represents the user's total app usage and is highly predictive of total screen time.
3. Gaming_App_Usage_Hours – High gaming app engagement significantly increasing screen time.
4. Social_Media_Usage_Hours – The time spent on social media is another major driver of daily device use.
5. User_ID – This feature appears influential, but including it as a predictor is generally not meaningful, since it is just an identifier and not a true behavioral or demographic variable. Its presence here suggests it should be removed in future modeling for valid interpretation.

Comparison: SHAP vs. Model Feature Importance

- The Random Forest feature importances and the SHAP values both highlight Productivity, Total usage, Gaming, and Social Media hours as the main predictive drivers of the target variable.
- Minor discrepancies may exist in the ranking of features, but the overall interpretation is consistent: app usage categories dominate prediction, while demographic/location features have much less influence.
- The appearance of User_ID in the top rankings by both methods signals a data/modeling issue—random IDs, unless meaningful, should not guide predictions.

Domain Meaningfulness

- The results are logical and meaningful for the digital wellbeing/mobile behavior analysis domain:
 - Individuals spending more time on productivity, gaming, and social media apps naturally have higher daily screen times.
 - Age and number of apps used are somewhat important, suggesting behavioral differences across age groups or users with different app engagement patterns.
 - Location and Gender have only minor influence—which aligns with the intuition that behavioral usage patterns are more individualized than group-based at this level.
- The inclusion of User_ID as important is not meaningful; in valid analytic practice, unique identifiers should be excluded to prevent data leakage or model overfitting to spurious patterns.

Summary: Productivity, total usage, gaming, and social media app engagement are the dominant drivers of daily screen time—an outcome that aligns with expectations in the context of mobile digital behavior. Both SHAP and Random Forest feature importance support this conclusion, and future analysis should focus on meaningful behavioral and demographic features only.

